

The mediating effect of motivational types in the relationship between perfectionism and academic burnout



Eunbi Chang, Ahram Lee, Eunji Byeon, Hyunmo Seong, Sang Min Lee *

Korea University

ARTICLE INFO

Article history:

Received 27 July 2015

Received in revised form 30 September 2015

Accepted 3 October 2015

Available online 22 October 2015

Keywords:

Academic burnout

Perfectionism

Motivation

Korean undergraduates

ABSTRACT

This study examines the mediating role of motivation in the relationship between perfectionism and academic burnout in Korean undergraduates. To measure perfectionism, two types of scales (i.e., APS-R and HFMPs) were used. Also, five types of motivation (i.e., amotivation, external motivation, introjected motivation, identified motivation, and intrinsic motivation) were examined, distinctively. As a result, intrinsic motivation mediated the relationship between adaptive perfectionism, namely high standards and self-oriented perfectionism, and academic burnout. Identified motivation showed the same results as intrinsic motivation. That is, adaptive perfectionism was positively associated with greater levels of both intrinsic and identified motivation and, in turn, greater intrinsic (or identified) motivation was negatively associated with academic burnout. Meanwhile, there were mediated effects of amotivation in the relationship between maladaptive perfectionism (i.e., discrepancy and socially prescribed perfectionism) and academic burnout. Specifically, maladaptive perfectionism was positively related to greater level of amotivation and, in turn, greater amotivation was positively related with academic burnout. Lastly, introjected motivation only mediated the link between socially prescribed perfectionism and academic burnout. The practical implications were discussed.

© 2015 Elsevier Ltd. All rights reserved.

Research on academic burnout has gained importance in recent years. Studies on burnout were initially conducted to examine the chronic stress of service providers, such as nurses and social workers (Maslach & Schaufeli, 1993). This limited attention to service providers was mainly due to the fact that the major measurement of burnout, the Maslach Burnout Inventory (MBI), had dimensions that reflected interactions with the recipients of services (Schaufeli, Martinez, Marques-Pinto, Salanova, & Bakker, 2002). Later, the publication of a more generalized version, the Maslach Burnout Inventory - General Survey (MBI-GS), led to the expansion of this area of study. Furthermore, the Maslach Burnout Inventory - Student Survey (MBI-SS) was developed to assess academic burnout in university students (Schaufeli et al., 2002). Since then, the concept of burnout is widely employed among students (Bakker, Demerouti, & Schaufeli, 2002; Demerouti, Bakker, Nachreiner, & Schaufeli, 2001; Leiter & Schaufeli, 1996). There are three symptoms of academic burnout, identified by Schaufeli et al. (2002), which are emotional exhaustion, cynicism, and incompetence; all of these are caused by heavy study demands, course loads, and chronic stress. Emotional exhaustion is associated with loss of physical or emotional energy, due to study demands (Shin, Puig, Lee, Lee, & Lee, 2011). Cynicism relates to the students' feelings of indifference or disinterest towards academic activities (Shin et al., 2011). Lastly,

incompetence indicates the sense of reduced accomplishment, leading students to produce poorer academic achievements (Shin et al., 2011).

There are numerous studies examining academic burnout in university students. Gan, Shang, and Zhang (2007) examined academic burnout in Chinese university students, and found that their coping flexibility negatively predicted burnout. Salanova, Schaufeli, Martinez, and Bresó (2010) found that burnout was significantly associated with the presence of obstacles in academic performance, and the absence of the facilitators of performance among university students. Jacobs and Dodd (2003) showed that a negative temperament and subjective workload of college students predicted a high level of burnout.

As such, many variables related to academic burnout have been the focus of burnout research. One of the variables that this study intends to examine is perfectionism. Perfectionism is a multidimensional construct that has been viewed as a personality trait (Miquelon, Vallerand, Cardinal, & Grouzet, 2005). Numerous studies have tried to identify the different dimensions of perfectionism; according to Hewitt and Flett (1991), perfectionism can be distinguished into three dimensions, namely self-oriented perfectionism, other-oriented perfectionism, and socially prescribed perfectionism. Self-oriented perfectionism indicates an individual's setting of high goals and expectations for her or himself, and striving to achieve them (Hewitt & Flett, 1991). An individual with other-oriented perfectionistic traits expects others to perform perfectly, so as to meet her or his own high standards, and constantly evaluates others (Hewitt & Flett, 1991). Socially prescribed perfectionism is based on the expectations that are prescribed by significant others. An

* Corresponding author at: Department of Education, College of Education, Korea University, Anam-dong, Seongbuk-gu, Seoul, Korea.
E-mail address: leesang@korea.ac.kr (S.M. Lee).

individual with socially prescribed perfectionism is sensitive to the goals imposed by these significant others, and tries to achieve those goals in order to avoid their disapproval (Hewitt & Flett, 1991). Among these three dimensions, proposed by Hewitt and Flett (1991), self-oriented perfectionism and socially prescribed perfectionism are the dimensions that indicate the perfectionistic expectations of one's own self rather than of others (Chang & Rand, 2000; Miquelon et al., 2005). Thus, this study excludes other-oriented perfectionism, and only examines self-oriented perfectionism and socially prescribed perfectionism.

Another, more general distinction of perfectionism is the differentiation between adaptive and maladaptive perfectionism. Adaptive perfectionism is related to one's striving to perform and achieve better, while maladaptive perfectionism leads to worrying about being evaluated and judged (Enns, Cox, Sareen, & Freeman, 2001). According to some studies (e.g., Frost, Heimberg, Holt, Mattia, & Neubauer, 1993; Stoeber & Otto, 2006), adaptive perfectionism can be also called as perfectionistic striving (or positive striving) and maladaptive perfectionism can be replaced by perfectionistic concern (or maladaptive evaluation concerns).

According to previous studies, socially prescribed perfectionism is considered to be maladaptive, because the expectations imposed by significant others are perceived to be excessive and uncontrollable, leading to negative psychological adjustments (Bieling, Israeli, Smith, & Antony, 2003; Chang & Rand, 2000; Hewitt & Flett, 1991; Miquelon et al., 2005). On the other hand, self-oriented perfectionism can either be adaptive or maladaptive. According to Hewitt and Flett (1991), self-oriented perfectionism is associated with self-criticism and self-blame. However, it can also result in positive psychological adjustment, because the goals and standards set by oneself are perceived to be under one's control (Flett, Hewitt, Blankstein, & Dynin, 1994). Thus, self-oriented perfectionism can be considered to be a more adaptive form of perfectionism (Bieling et al., 2003; Frost et al., 1993; Miquelon et al., 2005; Suddarth & Slaney, 2001) and classified as a perfectionistic strivings (Hill & Curran, 2015; Stoeber & Otto, 2006).

Since there is extensive research on identifying different dimensions of perfectionism, it would be important to clarify the constructs being examined in the study. In a previous study that examined the relationship between perfectionism and academic burnout, Chang, Lee, Byeon, and Lee (2015) used a Multidimensional Perfectionism Scale (HFMPs; Hewitt & Flett, 1991) to measure self-oriented and socially prescribed perfectionism. However, this study was limited in that the wide scope of perfectionism was measured using only one scale.

It was suggested that various measures could be used to investigate perfectionism. In addition, Slaney, Rice, Mobley, and Trippi (2001) raise the question of whether previously used scales- regarding perfectionism- are really measuring the essence of perfectionism itself. Thus, Slaney et al. (2001) developed a revised version of the Almost Perfect Scale (APS-R) in order to capture the common definition of perfectionism, as well as both the positive and negative aspects of perfectionism. The APS-R consists of three subscales, namely high standards, order, and discrepancy. The high standards subscale measures whether one has high expectations for one's own performance. The order subscale is associated with whether one prefers orderliness, and the discrepancy subscale is related to the perceived gap between one's performances and standards (Rice, Ashby, & Slaney, 2007; Slaney et al., 2001). Among these three, the high standards subscale is used to assess the adaptive form of perfectionism while the discrepancy scale measures the maladaptive form of perfectionism (Rice & Slaney, 2002; Rice, Richardson, & Tueller, 2014; Slaney et al., 2001; Suddarth & Slaney, 2001). Many researchers (e.g., Rice et al., 2014; Stoeber & Otto, 2006) have insisted that high standards and discrepancy are the main factors of the APS-R, so we only considered these two factors in this study.

To measure perfectionism, this study used APS-R and HFMPs concurrently. While the APS-R clearly distinguishes between adaptive

and maladaptive perfectionism, the self-oriented perfectionism scale of HFMPs has been found to be either adaptive or maladaptive. However, since self-oriented perfectionism is viewed as a more adaptive form in many studies (e.g., Cox, Enns, & Clara, 2002; Flett et al., 1994; Frost et al., 1993; Miquelon et al., 2005; Zhang, Gan, & Chan, 2007), examining both the self-oriented perfectionism scale of HFMPs and the high standards scale of the APS-R would provide a clearer view of adaptive perfectionism (i.e., perfectionistic strivings). Some meta-studies (Hill & Curran, 2015; Stoeber & Otto, 2006) also distinguished these two kinds of perfectionism as perfectionistic strivings. On the other hand, the socially prescribed perfectionism scale of the HFMPs along with discrepancy scale of the APS-R would measure maladaptive perfectionism (i.e., perfectionistic concerns) as indicated in the previous studies (Hill & Curran, 2015; Stoeber & Otto, 2006).

In academic settings, adaptive perfectionism is associated with academic engagement, while maladaptive perfectionism is related to academic burnout (Zhang et al., 2007; Jo & Lee, 2010). Students with adaptive perfectionism set their own goals, and strive to achieve them, by facilitating motivation and increasing the level of their performance (Jo & Lee, 2010). Students with maladaptive perfectionism, on the other hand, may set unrealistic goals due to external expectations and standards, and force themselves to perform in an exceedingly competitive manner, eventually leading to academic burnout (Shim, 1995). A previous study conducted by Chang et al. (2015) attempted to identify the relationship between perfectionistic traits and symptoms of academic burnout, and found that motivation was a key mediating factor. That is, intrinsic motivation partially mediated the relationship between self-oriented perfectionism and academic burnout, while extrinsic motivation fully mediated the relationship between socially prescribed perfectionism and academic burnout (Chang et al., 2015). This study was meaningful, in that it showed the path by which these perfectionistic traits influence academic burnout, and the importance of motivation as a mediator. However, Chang et al. (2015) did not consider the different dimensions of motivation, but rather divided motivation- dichotomously- into intrinsic and extrinsic.

There are extensive branches of theories regarding motivation, and each theory conceptualizes motivation in a different way. It would be essential to clearly identify the theoretical framework of motivation being examined. In this study, the self-determination theory (SDT) is applied, to understand the different dimensions of motivation. The SDT explains that motivation is related to the regulation of behaviors across one's life span, and it can largely be divided into intrinsic motivation, extrinsic motivation, and amotivation (Ryan & Deci, 2000). Intrinsic motivation refers to the inherent tendency to venture for one's own enjoyment and satisfaction, while amotivation indicates one's unwillingness to act (Ryan & Deci, 2000). Extrinsic motivation is the attainment of separable outcomes, based on social pressure and norms, and it consists of different degrees of self-determination and autonomy, ranging from external regulation to introjected regulation, identified regulation, and integrated regulation (Ryan & Deci, 2000).

Among these four extrinsic motivation types, external regulation and introjected regulation have a more external locus of causality, that is, behaviors based on these regulations are less autonomous and performed to acquire external rewards (Ryan & Deci, 2000). Identified and integrated regulations are also forms of extrinsic motivation, but they have a more internal locus of causality (Ryan & Deci, 2000). In other words, when there is social pressure for individuals to perform certain behaviors, the individuals take in the social value and transform it into their own (Ryan & Deci, 2000). The characteristic of identified regulation and integrated regulation would be the internalization of the social values, and such internalization leads these extrinsic motivational types to be more self-determined, functioning more similarly as intrinsic motivation (Ryan & Deci, 2000). Thus, some studies combine external and introjected forms of regulation as a controlled motivation composite, while combining identified, integrated, and intrinsic forms of regulations as an autonomous motivation composite (e.g., Williams,

Grow, Freedman, Ryan, & Deci, 1996). However, the distinction of controlled and autonomous does not capture all the degrees of motivation proposed by the SDT.

Motivation is in a continuum, making it difficult to distinguish one form from another, and the four types of motivation (namely intrinsic, identified, introjected, and external) are highly correlated (e.g., Chang et al., 2015; Stoeber, Feast, & Hayward, 2009). However, it would still be meaningful to examine the mediating effect of all of the different motivational types classified by the SDT separately since the previous studies (e.g., Vansteenkiste, Smeets, Soenens, Lens, Matos, & Deci, 2010) have mainly focused on the autonomous and controlled composites. Therefore, this study attempts to expand upon the previous research by examining the mediating effect of amotivation, intrinsic motivation, and the three types of extrinsic motivation (i.e., external, introjected, identified) on the relationship between perfectionism and academic burnout.

Specifically, intrinsic motivation and identified motivation have been reported to be positively related to adaptive perfectionism (e.g., Miquelon et al., 2005; Stoeber et al., 2009). Thus, these two types would be considered in the relationship between adaptive perfectionism, namely self-oriented perfectionism and high standards. On the other hand, introjected motivation, external motivation and amotivation have been reported to have positive associations with maladaptive perfectionism (e.g., Miquelon et al., 2005; Stoeber et al., 2009). When individuals consider themselves to be incompetent or think they cannot get valuable outcome because of low contingency between behaviors and outcomes, they are amotivated (Vallerand & Bissonnette, 1992; Vallerand & Ratelle, 2002). Behaviors of amotivated individuals are often regarded as being caused by outside forces (Vallerand & Ratelle, 2002). Viewed in this light, people who experience the discrepancy between goals and actual performance can be amotivated. Also, since goals of individuals with socially prescribed perfectionism are imposed by others and generally perceived as being difficult to achieve (Hewitt & Flett, 1991), socially prescribed perfectionists can be easily amotivated. Therefore, this study examined introjected motivation, external motivation, and amotivation in relation to maladaptive perfectionism, which is socially prescribed perfectionism and discrepancy.

The aim of the current study was to examine the relationships between academic burnout and the two different dimensions of perfectionistic traits— that is, adaptive perfectionism (i.e., self-oriented perfectionism, high standards) and maladaptive perfectionism (i.e., socially prescribed perfectionism, discrepancy). Also, the mediating effect of motivation was examined. It was hypothesized that intrinsic and identified motivation have mediating effects on the negative relationship between adaptive perfectionism (i.e., self-oriented perfectionism, high standards) and academic burnout, while external and introjected motivation, as well as amotivation, have mediating effects on the positive relationship between maladaptive perfectionism (i.e., socially prescribed perfectionism, discrepancy) and academic burnout.

1. Method

1.1. Participants

A total of 345 undergraduate students, aged from 19 to 29, participated in this study. The participants were recruited from a general education course at a university located in Seoul. The participants completed the survey on their own time, and at a location available to them, using the online survey tool SurveyMonkey (SurveyMonkey.com, LLC). There were 213 males (62%) and 132 females (38%); 18.3% of these self-identified as freshmen, 48.1% were sophomores, 21.7% were juniors, and 11.9% were seniors. The participants were offered some extra points for their course as an incentive for completing the survey.

1.2. Measures

1.2.1. Academic Burnout

The Korean version of the MBI-SS (Shin et al., 2011) was used to measure the academic burnout levels of participants. Shin et al. (2011) examined the factorial validity of the Korean version of MBI-SS. They confirmed that the three factor model of burnout is more appropriate compared to one factor model. Also, they found that factor loadings were equivalent regardless of academic setting (i.e., middle and high school) and gender (i.e., male and female) using multi group analyses. The MBI-SS consists of three dimensions of academic burnout: emotional exhaustion (EE), cynicism (CY), and academic inefficacy (AI). This self-reported scale is comprised of fifteen items: five of which measure emotional exhaustion (e.g., “Studying or attending a class is really a strain for me”), four of which measure cynicism (e.g., “I have become less enthusiastic about my studies”), and six of which measure efficacy (e.g., “I can effectively solve the problems that arise in my studies”). Academic efficacy was reverse-scored, to be used in the final analysis as academic inefficacy. The participants responded to each item using a scale of 1 (strongly disagree) through 5 (strongly agree). Shin et al. (2011) reported reliability estimates for the MBI-SS as .86 for exhaustion, .82 for cynicism, and .82 for academic inefficacy. In the current study, the alpha coefficients were .87 for the total MBI-SS, and .81, .86, and .80 for exhaustion, cynicism, and academic inefficacy, respectively.

1.2.2. Perfectionism

Perfectionism was measured using Hewitt and Flett (1991)'s Multidimensional Perfectionism Scale (HFMPs) and the Almost Perfect Scale-Revised (APS-R), by Slaney et al. (2001). In this study, the Korean version of the HFMPs, a 30-item scale translated and cross-validated by Han (1993), was used to measure self-oriented and socially prescribed perfectionism. Han (1993) confirmed three factors of translated version of HFMPs using factor analysis. Also, they (Han, 1993) examined the relation between HFMPs and other measures such as Irrational Belief Test (IBT; Jones, 1968) and Dysfunctional Attitude Scale-Form A (DAS-A; Weissman & Beck, 1978). The DAS-A is classified into two factors, namely performance evaluation and approval by others (Cane, Olinger, Gotlib, & Kuiper, 1986). In Han (1993)'s study, both self-oriented perfectionism and socially prescribed perfectionism scales of HFMPs had significant positive correlation with two factors (i.e., performance evaluation, approval by others) of DAS-A. In addition, self-oriented perfectionism scale of HFMPs had positive correlation with high self-expectations scale and frustration reactive scale of IBT. Socially prescribed perfectionism scale of HFMPs was positively correlated with demand for approval scale and high self-expectations scale of IBT (Han, 1993). Fifteen items were used to measure self-oriented perfectionism (e.g., “I must work to my full potential at all times”), and another fifteen items measures socially prescribed perfectionism (e.g., “My family expects me to be perfect”). Each item was rated on a 7-point Likert scale (1 = strongly disagree, 7 = strongly agree). In this study, Cronbach's alphas were .88 for self-oriented perfectionism and .72 for socially prescribed perfectionism.

The APS-R was also used to measure adaptive and maladaptive aspects of perfectionism. In this study, we used the Korean translated version of the APS-R, which was translated by Kim (2005). There are 23 items, divided into three subscales: high standards (e.g., “I have a strong need to strive for excellence”), order (e.g., “I think things should be put away in their place”), and discrepancy (e.g., “Doing my best never seems to be enough”). Among these three, only high standards and discrepancy were used for analysis. The items for high standards measured adaptive perfectionism, and the items for discrepancy measured maladaptive perfectionism (Slaney et al., 2001). All items were responded to on a 7-point Likert-type scale ranging from 1 (strongly disagree) to 7 (strongly agree). In Kim (2005)'s study, there were positive correlation between high standards of APS-R and self-esteem. Also, discrepancy of APS-R had positive correlation with self-critical

depression and negative correlation with self-esteem. In this study, the alpha coefficients were .80 for high standards, .71 for order, and .84 for discrepancy.

1.2.3. Academic Motivation

The Academic Motivation Scale – College Version (AMS-C; Vallerand et al., 1992) is the English translation of the Échelle de Motivation en Éducation (EME; Vallerand, Blais, Brière, & Pelletier, 1989), which was developed to assess students' motivation types, as applied to academic activities. This 28-item scale is divided into seven subscales, reflecting three subscales for intrinsic motivation, three subscales for extrinsic motivation, and one subscale for amotivation. The three types for intrinsic motivation are: to know (e.g., “Because I experience pleasure and satisfaction while learning new things”); towards accomplishment (e.g., “For the pleasure I experience while surpassing myself in my studies”); and to experience stimulation (e.g., “For the pleasure that I experience when I read interesting authors”). The three types of extrinsic motivation are: external regulation (e.g., “Because with only a high-school degree I would not find a high-paying job later on”); introjected regulation (e.g., “Because I want to show myself that I can succeed in my studies”); and identified regulation (e.g., “Because I think that a college education will help me better prepare for the career I have chosen”). The last type of motivation is amotivation (e.g., “Honestly, I don't know; I really feel that I am wasting my time in school”). On the AMS-C, participants are asked “Why do you go to college?” and four possible responses are given for each of the motivation styles. Each item was answered on 7-point scales, ranging from 1 (does not correspond at all) to 7 (corresponds exactly).

The AMS-C was translated into Korean by the researchers who conducted this study. The AMS-C was first translated into Korean by a bilingual individual who did not participate in this study. The translated version was then back-translated by another bilingual individual who has not seen the original version of the AMS-C. Then, the researchers compared the back-translation and the original version of the AMS-C for any discrepancies. A few words on the Korean version were modified accordingly for clarification, and the translation and the back-translation process were both conducted again. The Korean version was finalized when the result was fully agreeable to every concerned member. The Cronbach's alphas of each subscale of the Korean translation of the AMS-C were as follows: .82 for to know, .82 for accomplishment, .79 for stimulation, .79 for external regulation, .80 for introjected regulation, .69 for identified regulation, and .89 for amotivation. Although the original version divided intrinsic motivation into three types, these three subscales were included as one intrinsic motivation factor ($\alpha = .90$) for the analysis.

1.3. Data Analysis.

IBM SPSS Statistics Developer Version 21.0 was used in order to analyze the given data, collected from a total of 345 students. First, descriptive statistics were used to examine the tendency of the variables. After that, reliability and correlation analyses were conducted to check and understand the internal consistency of, and relations among, the variables of perfectionism, motivation, and academic burnout.

Following this, mediation analyses were conducted. In this study, a bootstrapping approach has been used, because this approach would eliminate the assumption of the normality of the sampling distribution (Hayes, 2009; Mooney & Duval, 1993; Preacher & Hayes, 2004). The bootstrapping method, which includes drawing repeated data samples with replacement, has been found to yield bias-corrected confidence intervals (Preacher & Hayes, 2008). Moreover, according to Preacher and Hayes (2008), Type II errors are reduced, as the bootstrapping approach requires fewer inferential tests. To implement this approach, an SPSS macro process (Preacher, Rucker, & Hayes, 2007) was used. There were two estimation methods in the macro process, to produce bootstrapped percentile and normal distribution confidence intervals, for indirect effects. This macro program enables researchers to obtain

the total and specific indirect effects of the independent variable on the dependent variable.

In this study, a total of ten mediation models were tested. First, two mediation models were tested. That is, the mediating role of intrinsic motivation or identified motivation in the relationship between self-oriented perfectionism (IV) and academic burnout (DV) were tested, respectively. Also, three mediation models were analyzed. Specifically, the mediating role of identified motivation, external motivation or amotivation in the link between socially prescribed perfectionism (IV) and academic burnout (DV) were analyzed, respectively. Next, another two mediation models were tested. That is, intrinsic or identified motivation was considered as a mediator in the relationship between high standards (IV) and academic burnout (DV), respectively. Finally, three mediation models were analyzed; specifically, the mediating role of introjected motivation, external motivation, or amotivation in the relationship between discrepancy and academic burnout were tested, respectively.

2. Results

2.1. Descriptive and correlation analyses.

The means, standard deviations, and correlations among all the observed variables were presented in Table 1. First, intrinsic motivation was positively correlated with self-oriented perfectionism ($r = .29$, $p < .01$) and high standards ($r = .33$, $p < .01$), but not with socially prescribed perfectionism and discrepancy. Identified motivation showed a positive relationship with self-oriented perfectionism ($r = .22$, $p < .01$) and high standards ($r = .37$, $p < .01$), and a negative relationship with discrepancy ($r = -.19$, $p < .01$). There was no significant correlation between identified motivation and socially prescribed perfectionism. In the case of introjected motivation, there was a positive correlation with self-oriented perfectionism ($r = .23$, $p < .01$), high standards ($r = .34$, $p < .01$) and even with socially prescribed perfectionism ($r = .20$, $p < .01$), but not with discrepancy. Also, external motivation was positively correlated with self-oriented perfectionism ($r = .12$, $p < .05$), socially prescribed perfectionism ($r = .15$, $p < .01$) and high standards ($r = .19$, $p < .05$), respectively. Meanwhile, it was found that there was a negative correlation between amotivation and self-oriented perfectionism ($r = -.12$, $p < .05$). Amotivation also showed a positive correlation with socially prescribed perfectionism ($r = .13$, $p < .05$) and discrepancy ($r = .42$, $p < .01$).

Academic burnout had a positive correlation with external motivation ($r = .23$, $p < .01$), amotivation ($r = .57$, $p < .01$), socially prescribed perfectionism ($r = .20$, $p < .01$) and discrepancy ($r = .43$, $p < .01$), respectively. On the other hand, academic burnout was found to have negative correlations with intrinsic motivation ($r = -.34$, $p < .01$), identified motivation ($r = -.24$, $p < .01$), self-oriented perfectionism ($r = -.19$, $p < .01$) and high standards ($r = -.23$, $p < .01$). These results imply that the higher levels of external motivation, amotivation, socially prescribed perfectionism, and discrepancy were related to higher level of academic burnout. Meanwhile, when the scores of the level of intrinsic motivation, identified motivation, self-oriented perfectionism, and high standards increase, the level of academic burnout descends.

Lastly, high standards showed a higher correlation with self-oriented perfectionism ($r = .74$, $p < .01$) than socially prescribed perfectionism ($r = .16$, $p < .01$). On the other hand, discrepancy had positively significant relations with both self-oriented perfectionism ($r = .18$, $p < .01$) and socially prescribed perfectionism ($r = .37$, $p < .01$), but the correlation with socially prescribed perfectionism was higher.

2.2. Mediation analyses.

2.2.1. Mediating role of intrinsic/identified motivation between self-oriented perfectionism and academic burnout

In Fig. 1, the mediation model with intrinsic motivation (which mediates the relationship between self-oriented perfectionism and

Table 1

The means, standard deviations and subscale correlations.

	M	SD	1	2	3	4	5	6	7	8	9	10
1. Academic burnout	2.84	.58	–									
2. Intrinsic motivation	4.09	.88	–.34**	–								
3. Identified motivation	5.23	.86	–.24**	.36**	–							
4. Introjected motivation	4.22	1.11	–.05	.48**	.31**	–						
5. External motivation	5.30	1.06	.23**	–.05	.43**	.38**	–					
6. Amotivation	3.21	1.33	.57**	–.32**	–.43**	–.11*	.00	–				
7. Self-oriented perfectionism	4.44	.75	–.19**	.29**	.22**	.23**	.12*	–.12*	–			
8. Socially prescribed perfectionism	3.85	.52	.20**	.06	.00	.20**	.15**	.13*	.36**	–		
9. High standards	4.73	.79	–.23**	.33**	.37**	.34**	.19*	–.14	.74**	.16**	–	
10. Discrepancy	3.85	.77	.43**	–.03	–.19**	.08	.07	.42**	.18**	.37**	.24**	–

Note. * $p < .05$, ** $p < .01$.

academic burnout) is shown. The total effect of self-oriented perfectionism on academic burnout was significant ($t = -3.60, p < .001$; c path). Also there was a significant effect of self-oriented perfectionism on intrinsic motivation ($t = 5.52, p < .001$; a path) and of intrinsic motivation on academic burnout ($t = -5.88, p < .001$; b path), based on unstandardized regression coefficients. Since the bootstrapped 95% confidence interval did not contain zero ($-.11, -.03$), the indirect effect was significant. Furthermore, the main effect of self-oriented perfectionism on academic burnout was not significant, after controlling for the effect of intrinsic motivation ($t = -1.93, p = .054$; c' path), suggesting full mediation.

The mediating role of identified motivation between self-oriented perfectionism and academic burnout is also presented in Fig. 1. There was a significant effect of self-oriented perfectionism on identified motivation ($t = 4.18, p < .001$; a path) and of identified motivation on academic burnout ($t = -3.98, p < .001$; b path). The indirect effect of self-oriented perfectionism on academic burnout was significant (95% CI: $-.07, -.01$), but the main effect was also significant when the effect of identified motivation was controlled ($t = -2.71, p < .01$; c' path). This means that identified motivation partially mediated the relationship between self-oriented perfectionism and academic burnout.

2.2.2. Mediating role of intrinsic/identified motivation between high standards and academic burnout

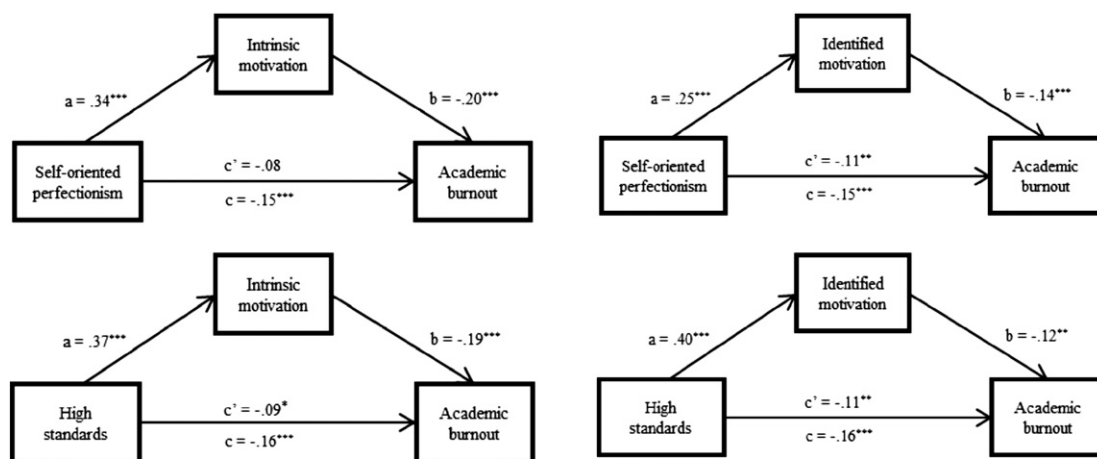
Fig. 1 represents the mediation model of intrinsic (or identified) motivation in the relationship between high standards and academic burnout. There was a significant total effect of high standards on

academic burnout ($t = -4.27, p < .001$; c path). The effects of high standards on intrinsic motivation ($t = 6.43, p < .001$; a path), and of intrinsic motivation on academic burnout ($t = -5.58, p < .001$; b path) were significant. A significant indirect effect was found, based on the bootstrapped 95% confidence interval, which did not contain zero ($-.11, -.04$). Moreover, the main effect of high standards on academic burnout was significant when controlling for the effect of intrinsic motivation ($t = -2.38, p < .05$; c' path), which means partial mediation.

Also, high standards positively affected identified motivation ($t = 7.36, p < .001$; a path), and identified motivation had a negative effect on academic burnout ($t = -3.33, p < .01$; b path). The indirect effect was confirmed to be significant, by a 95% confidence interval of bootstrapping ($-.09, -.02$). Since the direct effect of high standards on academic burnout was significant when the identified motivation effect was controlled ($t = -2.79, p < .01$; c' path), identified motivation was considered to be a partial mediator.

2.2.3. Mediating role of introjected/external/amotivation between socially prescribed perfectionism and academic burnout

Since introjected motivation did not affect academic burnout significantly ($t = -1.61, p = .11$; b path), the mediation model was not tested. The result of the mediation model with external motivation (or amotivation) mediating the relationship between socially prescribed perfectionism and academic burnout was shown in Fig. 2. A significant total effect of socially prescribed perfectionism on academic burnout was found ($t = 3.74, p < .001$; c path). Both the effects of socially prescribed perfectionism on external motivation ($t = 2.85, p < .01$; a path)

Note. Regression coefficients are unstandardized, * $p < .05$, ** $p < .01$, *** $p < .001$ **Fig. 1.** A mediation model of intrinsic/identified motivation in the relationship between adaptive perfectionism and academic burnout.

and external motivation on academic burnout ($t = 3.93, p < .001$; b path) were significant. Additionally, the indirect effect of socially prescribed perfectionism on academic burnout, through external motivation, had a 95% confidence interval between .01 and .07, which indicates a significant mediation effect. Even after the effect of external motivation was controlled, the main effect of socially prescribed perfectionism on academic burnout was still significant ($t = 3.18, p < .01$; c' path). This means that the external motivation partially mediated the relationship between socially prescribed perfectionism and academic burnout.

In addition, socially prescribed perfectionism had a significant effect on amotivation ($t = 2.44, p < .05$; a path), and amotivation showed a significant effect on academic burnout ($t = 12.54, p < .001$; b path) as well. The indirect effect of socially prescribed perfectionism on academic burnout was significant, since the bootstrapped 95% confidence interval did not contain zero (.01, .15). However, when the effect of amotivation was controlled, the main effect was still significant ($t = 2.84, p < .01$; c' path), indicating partial mediation.

2.2.4. Mediating role of introjected/external/amotivation between discrepancy and academic burnout

Since each effect of discrepancy on external motivation ($t = 1.22, p = .22$; a path) and introjected motivation ($t = 1.49, p = .14$; a path) was not significant, both mediation models with external motivation and introjected motivation could not be tested. However, as shown in Fig. 2, the mediation model with amotivation, which mediates the relationship between discrepancy and academic burnout. The total effect of discrepancy on academic burnout was significant ($t = 8.80, p < .001$; c path). The effect of discrepancy on amotivation ($t = 8.56, p < .001$; a path), as well as that of amotivation on academic burnout ($t = 10.04, p < .001$; b path) were significant. Indeed, the significant indirect effect of discrepancy on academic burnout, through amotivation, was found; it had a 95% confidence interval between .11 and .20, meaning a significant mediation effect. After the effect of amotivation was controlled, the main effect of discrepancy on academic burnout was significant ($t = 4.86, p < .001$; c' path). This indicates that amotivation partially mediated the relationship between discrepancy and academic burnout.

3. Discussion

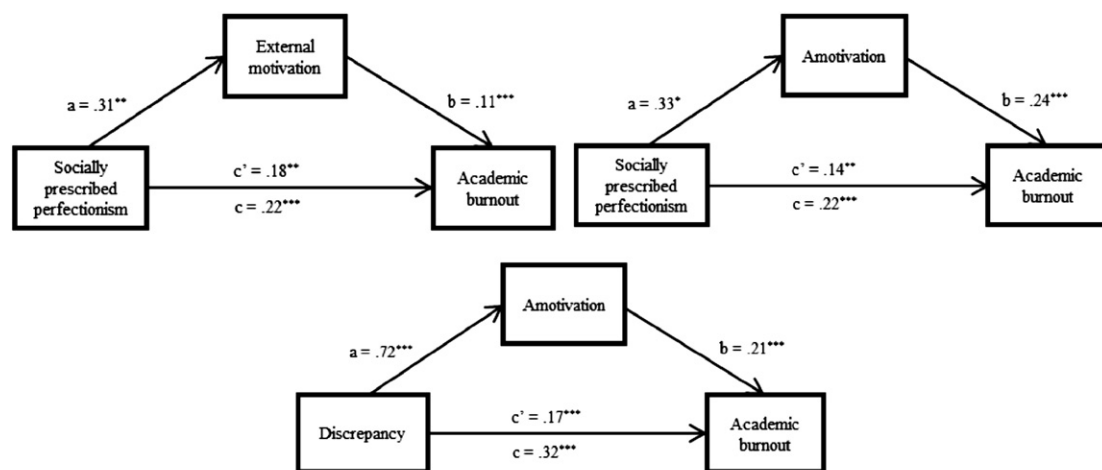
This study investigated the role of motivation as a mediator between perfectionism and academic burnout in undergraduate students. It attempted to use the mediation approach to identify the mechanism that could be the focus of interventions (Blankstein, Dunkley, & Wilson,

2008) to reduce or prevent academic burnout. Chang et al. (2015) attempted to identify these relationships in high school students. However, they did not consider the various types of perfectionism scales, except for the Multidimensional Perfectionism Scale (HFMP; Hewitt & Flett, 1991). In addition, they only explored intrinsic and external motivation, among the five types of motivation (amotivation, external motivation, introjected motivation, identified motivation, and intrinsic motivation) to identify the relationship between perfectionism, motivation, and academic burnout. In this regard, this study was executed to overcome the limitations of Chang et al.'s (2015) study, with undergraduate students as participants.

In this study, we measured perfectionism using not only the Multidimensional Perfectionism Scale (HFMP; Hewitt & Flett, 1991) but also the Almost Perfect Scale (APS-R; Slaney et al., 2001) to examine both adaptive and maladaptive perfectionism. Also, the role of the five types of motivation was distinctively explored in the relationship between perfectionism and academic burnout. The main findings of the research were consistent with that of Chang et al. (2015).

First, we confirmed the positive association between the self-oriented perfectionism scale of the HFMP, and the high standards scale of the APS-R. Although self-orientated perfectionism was positively correlated with both discrepancy and high standards, the correlation was stronger for high standards compared to discrepancy. Also, socially prescribed perfectionism had a high correlation with discrepancy, compared to high standards. These results are consistent with the previous studies (e.g., Dunkley, Blankstein, & Berg, 2012; Suddarth & Slaney, 2001), which showed that socially prescribed perfectionism is highly correlated with discrepancy, and self-oriented perfectionism is highly correlated with high standards.

Next, there are significant mediating effects of intrinsic and identified motivation in the relationship between self-oriented perfectionism and academic burnout, as well as high standards and academic burnout. That is, adaptive perfectionism (i.e., perfectionistic strivings), namely self-oriented perfectionism and high standards, affected both intrinsic motivation and identified motivation positively, and these motivation types had a negative effect on academic burnout, respectively. The significant association between adaptive perfectionism and intrinsic (or identified) motivation in this study are consistent with previous studies (e.g., Chang et al., 2015; Miquelon et al., 2005; Stoeber & Eismann, 2007). By definition, intrinsically motivated people engage in activities for fun and enjoyment and identified motivated individuals pursue an activity because it has meaning in connection with their goals and identities (Koestner & Losier, 2002). Thus, it seems that individuals with adaptive perfectionism may not experience academic burnout



Note. Regression coefficients are unstandardized, * $p < .05$, ** $p < .01$, *** $p < .001$

Fig. 2. A mediation model of external motivation/amotivation in the relationship between maladaptive perfectionism and academic burnout.

because their high standards reflect their inherent interest or perceived importance of study.

Previous studies (e.g., Guay & Vallerand, 1997; Ratelle, Guay, Vallerand, Larose, & Senécal, 2007; Ryan & Connell, 1989; Walls & Little, 2005) showed that intrinsic and identified motivation leads to positive outcomes in the academic domain. For example, Ratelle et al. (2007) reported that students with high autonomous motivation show high academic persistence. Intrinsic and identified motivation had a positive effect on the school adjustment (Walls & Little, 2005) and the continued enrollment in college (Koestner et al., as cited in Koestner & Losier, 2002). Also, identified motivation had significant negative association with psychological distress in college students (Koestner et al., as cited in Koestner & Losier, 2002). The present study can be interpreted in the same line with these previous studies.

In addition, considering the present results, which showed that self-oriented perfectionism and high standards are negatively related to academic burnout, it can be assumed that high standards alone does not cause a negative effect. That is, unless they experience a discrepancy between their goals and reality, they would not be faced with burnout symptoms. Zhang et al. (2007) confirmed in their study that the high standards which individuals set for themselves do not cause burnout. Thus, the factor that is more important in preventing burnout may be flexibility, in regard to being able readjust the preliminary goal, depending on one's own ability (Zhang et al., 2007).

Meanwhile, amotivation and external motivation partially mediated the relationship between socially prescribed perfectionism and academic burnout. Specifically, socially prescribed perfectionism positively related with both amotivation and external motivation, which affected academic burnout positively. Amotivation is a state of having no reason to make an effort to learn or achieve something (Cheon & Reeve, 2015), and originally contrasted with other motivation types within the self-determination theory (Ryan & Deci, 2000). It occurs when people feel incompetent in doing something and cannot expect desired outcome (Ryan & Deci, 2000). In the same line with this logic, students who experience discrepancy may experience academic burnout because they would be in an amotivated state when they realize that it would be difficult for them to gain what they had expected. On the other hand, people with socially prescribed perfectionism will experience amotivation, and thus academic burnout, because they would study without any personal reasons or purpose.

In the case of external motivation, it mediated the link between socially prescribed perfectionism and academic burnout, but not the relationship between discrepancy and academic burnout. Although discrepancy is similar to socially prescribed perfectionism, in terms of reflecting the maladaptive aspects of perfectionism, these two concepts are still distinguishable. Discrepancy premises the goals set by oneself, and occurs when the gap one perceives between these goals and the actual states are large (Slaney et al., 2001). Thus, it seems that discrepancy is not related with introjected or external motivation, because it is associated with self-set goals, and not externally provided standards.

According to previous studies, maladaptive perfectionism (i.e., socially prescribed perfectionism and discrepancy) is related with negative outcomes, including hopelessness (Chang & Rand, 2000), psychological symptoms (Chang & Rand, 2000), self-blame about failure (Rice & Ashby, 2007), suicide ideation (Hewitt, Flett, & Weber, 1994), burnout (Mitchelson & Burns, 1998), and test anxiety (Mills & Blankstein, 2000; Stoeber et al., 2009). Otherwise, adaptive perfectionism (i.e., self-oriented perfectionism, high standards) is associated with both positive (e.g., Chang et al., 2015; Mills & Blankstein, 2000; Trumpeter, Watson, & O'Leary, 2006) and negative outcomes (e.g., Chang & Sanna, 2001; Hewitt et al., 1994); however, the current study showed a positive outcome, namely decreased academic burnout in undergraduate students, through intrinsic and identified motivation. These results are in alignment with Chang et al.'s (2015) recent study and supported by Hill and Curran (2015)'s study. In Hill and Curran (2015)'s meta-analysis, perfectionistic strivings (i.e., adaptive

perfectionism) had negative association with burnout, but perfectionistic concerns (i.e., maladaptive perfectionism) had positive relation with burnout. Also, the negative effect of perfectionistic strivings on burnout became larger after controlling the relation between two dimensions of perfectionism (i.e., perfectionistic strivings, perfectionistic concerns), which implied that pure perfectionistic strivings may prevent burnout as we found in this study.

The findings of this study have several practical implications. First, since perfectionism is a multidimensional construct, counselors or professors who meet students with perfectionistic traits should intervene differently, depending on the type of perfectionism. They should focus on the students with negative or maladaptive perfectionism, rather than positive or adaptive perfectionism. Also, they should pay attention and monitor the motivation styles of the students. In essence, promoting intrinsic motivation in students can be beneficial, eventually leading to lessening or even preventing academic burnout.

To address the motivation of students, Lepper and Hodell (1989) suggest that four factors (challenge, curiosity, control, and fantasy) should be considered when trying to improve student's intrinsic motivation. Specifically, tasks with intermediated difficulty are good for one's challenge, which can lead to intrinsic motivation (Schunk, Meece, & Pintrich, 2014). Also, curiosity can be stimulated by incongruent and surprising ideas (Schunk et al., 2014), and a sense of control can be enhanced by providing choices and an autonomous environment for students (Zuckerman, Porac, Lathin, Smith, & Deci, 1978). Making students experience fantasy, using simulations such as computer games, also can promote intrinsic motivation (Parker & Lepper, 1992; Lepper & Hodell, 1989). However, rewards including prize and money (Deci, 1972; Lepper, Greene, & Nisbett, 1973), punishment (Deci, 1972), negative feedback (Deci, 1972), and competition (Deci, Betley, Kahle, Abrams, & Porac, 1981) can decrease intrinsic motivation. Therefore, professors should try to avoid giving rewards, negative feedback, and making excessively competitive situations that could hinder the intrinsic motivation. Also, they should use various methods to promote students' intrinsic motivation.

In addition, for students with maladaptive perfectionism, a supportive attitude of autonomy from the teachers would be helpful (Cheon & Reeve, 2015; Reeve, 2009). Students experience amotivation when psychological needs are not satisfied (Deci & Ryan, 2000), and the controlling attitude of teachers frustrates the needs of students (Reeve, 2009). Reeve (2009) suggested several ways to become an autonomy-supportive teacher. For example, offering explanatory rationales about uninteresting tasks or activities, allowing enough time for self-paced learning, and accepting students' negative expressions would help teachers be more autonomy-supportive and less controlling. Although this suggestion was made for teachers of K to 12, this could be applied to faculty members at the university level, to encourage the autonomy of undergraduate students.

This study has several limitations. First, it is difficult to infer the causal relationship between the variables in the study, because we only investigated statistical mediation. Next, although we clearly distinguished the types of perfectionism and investigated each effect, they are inter-correlated with each other. Thus, in the future research, the effect of perfectionism should be investigated after controlling the correlation between each type of perfectionism. Third, it is difficult to generalize the results in other cultures. In collectivistic cultures, such as Korea, significant others influence an individual's life more compared to those in individualistic cultures, which can be inferred from previous studies (e.g., Bong, Hwang, Noh, & Kim, 2014; Markus & Kitayama, 1991). Therefore, replication of this study in other cultures should be executed in the future. Finally, the measures used in this study were developed in other countries, and were translated into Korean. Even though measures were translated and back-translated, they may reflect cultural biases. Thus, in the future studies, measures developed in Korea should be used.

Despite these limitations, this study extends previous studies (e.g., Chang et al., 2015) by examining five types of motivation

distinctively and using two different types of perfectionism scales. Also, this study reconfirms the results of previous studies with the sample of undergraduate students. Specifically, through this study, we confirmed that adaptive perfectionism negatively affects academic burnout, while maladaptive perfectionism positively affects academic burnout. Also, motivation has a mediating role in the link between perfectionism and academic burnout. Based on these findings, counselors and faculty can promote motivation to prevent academic burnout for students with perfectionistic traits.

Acknowledgment

This work was supported by the National Research Foundation of Korea Grant funded by the Korean Government (NRF-2014S1A5B8060944).

References

- Bakker, A. B., Demerouti, E., & Schaufeli, W. B. (2002). Validation of the maslach burnout inventory - general survey: an internet study. *Anxiety, Stress and Coping*, 15, 245–260. <http://dx.doi.org/10.1080/1061580021000020716>.
- Bieling, P. J., Israeli, A. L., Smith, J., & Antony, M. M. (2003). Making the grade: the behavioral consequences of perfectionism in the classroom. *Personality and Individual Differences*, 35, 163–178. [http://dx.doi.org/10.1016/S0191-8869\(02\)00173-3](http://dx.doi.org/10.1016/S0191-8869(02)00173-3).
- Blankstein, K. R., Dunkley, D. M., & Wilson, J. (2008). Evaluative concerns and personal standards perfectionism: self-esteem as a mediator and moderator of relations with personal and academic needs and estimated GPA. *Current Psychology*, 27, 29–61. <http://dx.doi.org/10.1007/s12144-008-9022-1>.
- Bong, M., Hwang, A., Noh, A., & Kim, S. I. (2014). Perfectionism and motivation of adolescents in academic contexts. *Journal of Educational Psychology*, 106, 711–729. <http://psycnet.apa.org/doi/10.1037/a0035836>.
- Cane, D. B., Olinger, L. J., Gotlib, I. H., & Kuiper, N. A. (1986). Factor structure of the dysfunctional attitude scale in a student population. *Journal of Clinical Psychology*, 42, 207–309.
- Chang, E. C., & Rand, K. L. (2000). Perfectionism as a predictor of subsequent adjustment: evidence for a specific diathesis-stress mechanism among college students. *Journal of Counseling Psychology*, 47, 129–137. <http://dx.doi.org/10.1037/0022-0167.47.1.129>.
- Chang, E. C., & Sanna, L. J. (2001). Optimism, pessimism, and positive and negative affectivity in middle-aged adults: a test of a cognitive-affective model of psychological adjustment. *Psychology and aging*, 16, 524–531. <http://dx.doi.org/10.1037/0882-7974.16.3.524>.
- Chang, E., Lee, A., Byeon, E., & Lee, S. M. (2015). Role of motivation in the relation between perfectionism and academic burnout in Korean students. *Personality and Individual Differences*, 82, 221–226. <http://dx.doi.org/10.1016/j.paid.2015.03.027>.
- Cheon, S., & Reeve, J. (2015). A classroom-based intervention to help teachers decrease students' amotivation. *Contemporary Educational Psychology*, 40, 99–111. <http://dx.doi.org/10.1016/j.cedpsych.2014.06.004>.
- Cox, B. J., Enns, M. W., & Clara, I. P. (2002). The multidimensional structure of perfectionism in clinically distressed and college student samples. *Psychological Assessment*, 14, 365–373. <http://dx.doi.org/10.1037/1040-3590.14.3.365>.
- Deci, E. L. (1972). The effects of contingent and noncontingent rewards and controls on intrinsic motivation. *Organizational Behavior and Human Performance*, 8, 217–229. [http://dx.doi.org/10.1016/0030-5073\(72\)90047-5](http://dx.doi.org/10.1016/0030-5073(72)90047-5).
- Deci, E. L., & Ryan, R. M. (2000). The "what" and "why" of goal pursuits: human needs and the self-determination of behavior. *Psychological Inquiry*, 11, 227–268. http://dx.doi.org/10.1207/S15327965PLI1104_01.
- Deci, E. L., Betley, G., Kahle, J., Abrams, L., & Porac, J. (1981). When trying to win competition and intrinsic motivation. *Personality and Social Psychology Bulletin*, 7, 79–83. <http://dx.doi.org/10.1177/014616728171012>.
- Demerouti, E., Bakker, A. B., Nachreiner, F., & Schaufeli, W. B. (2001). The job demands-resources model of burnout. *Journal of Applied Psychology*, 86, 499–512. <http://dx.doi.org/10.1037/0021-9010.86.3.499>.
- Dunkley, D. M., Blankstein, K. R., & Berg, J. L. (2012). Perfectionism dimensions and the five factor model of personality. *European Journal of Personality*, 26, 233–244. <http://dx.doi.org/10.1002/per.829>.
- Enns, M. W., Cox, B. J., Sareen, J., & Freeman, P. (2001). Adaptive and maladaptive perfectionism in medical students: a longitudinal investigation. *Medical Education*, 35, 1034–1042. <http://dx.doi.org/10.1111/j.1365-2923.2001.01044.x>.
- Flett, G. L., Hewitt, P. L., Blankstein, K. R., & Dynin, C. B. (1994). Dimensions of perfectionism and type a behavior. *Personality and Individual Differences*, 16, 477–485. [http://dx.doi.org/10.1016/0191-8869\(94\)90073-6](http://dx.doi.org/10.1016/0191-8869(94)90073-6).
- Frost, R. O., Heimberg, R. G., Holt, C. S., Mattia, J. I., & Neubauer, A. L. (1993). A comparison of two measures of perfectionism. *Personality and Individual Differences*, 14, 119–126. [http://dx.doi.org/10.1016/0191-8869\(93\)90181-2](http://dx.doi.org/10.1016/0191-8869(93)90181-2).
- Gan, Y., Shang, J., & Zhang, Y. (2007). Coping flexibility and locus of control as predictors of burnout among Chinese college students. *Social Behavior and Personality: An International Journal*, 35(8), 1087–1098. <http://dx.doi.org/10.2224/sbp.2007.35.8.1087>.
- Guay, F., & Vallerand, R. J. (1997). Social context, students' motivation, and academic achievement: toward a process model. *Social Psychology of Education*, 1, 211–233. <http://dx.doi.org/10.1007/BF02339891>.
- Han, K. Y. (1993). *Multidimensional perfectionism: Concept, measurement and relevance to maladjustment (doctoral dissertation)*. Seoul: Korea University.
- Hayes, A. F. (2009). Beyond Baron and Kenny: statistical mediation analysis in the new millennium. *Communication Monographs*, 76, 408–420. <http://dx.doi.org/10.1080/03637750903310360>.
- Hewitt, P. L., & Flett, G. L. (1991). Perfectionism in the self and social contexts: conceptualization, assessment, and association with psychopathology. *Journal of Personality and Social Psychology*, 60, 456–470. <http://dx.doi.org/10.1037/0022-3514.60.3.456>.
- Hewitt, P. L., Flett, G. L., & Weber, C. (1994). Dimensions of perfectionism and suicide ideation. *Cognitive Therapy and Research*, 18, 439–460. <http://dx.doi.org/10.1007/BF02357753>.
- Hill, A. P., & Curran, T. (2015). Multidimensional perfectionism and burnout a meta-analysis. *Personality and Social Psychology Review*, 1–20. <http://dx.doi.org/10.1177/1088868315596286>.
- Jacobs, S. R., & Dodd, D. (2003). Student burnout as a function of personality, social support, and workload. *Journal of College Student Development*, 44, 291–303. <http://dx.doi.org/10.1353/csd.2003.0028>.
- Jo, H., & Lee, H. (2010). The mediating effects of achievement goals on the relationship among perfectionism, academic burnout and academic engagement. *Korean Journal of Youth Studies*, 17(17), 131–154.
- Jones, R. G. (1968). A factored measure of Ellis' irrational belief system, with personality and maladjustment correlates. (doctoral dissertation). Retrieved from <http://repositories.tdl.org/ttu-ir/handle/2346/18941>.
- Kim, S. (2005). *A study of the relationships between perfectionism tendency, depression, and self-esteem of the undergraduate students: focused on the APS-R (the Almost Perfectionism Scales-Revised)*. (Master's dissertation) Seoul: Yonsei University.
- Koestner, R., & Losier, G. F. (2002). Distinguishing three ways of being internally motivated: a closer look at introjection, identification, and intrinsic motivation. In E. L. Deci, & R. M. Ryan (Eds.), *Handbook of self-determination research* (pp. 101–121). Rochester, NY: University of Rochester Press.
- Leiter, M. P., & Schaufeli, W. B. (1996). Consistency of the burnout construct across occupations. *Anxiety, Stress, and Coping*, 9, 229–243. <http://dx.doi.org/10.1080/10615809608249404>.
- Lepper, M. R., & Hodell, M. (1989). Intrinsic motivation in the classroom. *Research on Motivation in Education*, 3, 73–105.
- Lepper, M. R., Greene, D., & Nisbett, R. E. (1973). Undermining children's intrinsic interest with extrinsic rewards: A test of the "overjustification" hypothesis. *Journal of Personality and Social Psychology*, 28, 129–137. <http://dx.doi.org/10.1037/h0035519>.
- Markus, H. R., & Kitayama, S. (1991). Culture and the self: Implications for cognition, emotion, and motivation. *Psychological review*, 98, 224–253. <http://dx.doi.org/10.1037/0033-295X.98.2.224>.
- Maslach, C., & Schaufeli, W. B. (1993). Historical and conceptual development of burnout. In W. B. Schaufeli, C. Maslach, & T. Marek Professional (Eds.), *Burnout: Recent developments in theory and research* (pp. 1–16). Washington, DC: Taylor & Francis.
- Mills, J. S., & Blankstein, K. R. (2000). Perfectionism, intrinsic vs extrinsic motivation, and motivated strategies for learning: a multidimensional analysis of university students. *Personality and Individual Differences*, 29, 1191–1204. [http://dx.doi.org/10.1016/S0191-8869\(00\)00003-9](http://dx.doi.org/10.1016/S0191-8869(00)00003-9).
- Miquelon, P., Vallerand, R. J., Cardinal, G., & Grouzet, F. M. E. (2005). Perfectionism, academic motivation, and psychological adjustment: an integrated model. *Personality and Social Psychology Bulletin*, 31, 913–921. <http://dx.doi.org/10.1177/0146167204272298>.
- Mitchelson, J. K., & Burns, L. R. (1998). Career mothers and perfectionism: Stress at work and at home. *Personality and Individual Differences*, 2, 477–485. [http://dx.doi.org/10.1016/S0191-8869\(98\)00069-5](http://dx.doi.org/10.1016/S0191-8869(98)00069-5).
- Mooney, C. Z., & Duval, R. D. (1993). *Bootstrapping: A nonparametric approach to statistical inference*. Newbury Park, CA: Sage.
- Parker, L. E., & Lepper, M. R. (1992). Effects of fantasy contexts on children's learning and motivation: making learning more fun. *Journal of Personality and Social Psychology*, 62, 625–633. <http://dx.doi.org/10.1037/0022-3514.62.4.625>.
- Preacher, K. J., & Hayes, A. F. (2004). SPSS and SAS procedures for estimating indirect effects in simple mediation models. *Behavior Research Methods, Instruments, & Computers*, 36, 717–731. <http://dx.doi.org/10.3758/BF03206553>.
- Preacher, K. J., & Hayes, A. F. (2008). Asymptotic and resampling strategies for assessing and comparing indirect effects in multiple mediator models. *Behavior Research Methods*, 40, 879–891. <http://dx.doi.org/10.3758/BRM.40.3.879>.
- Preacher, K. J., Rucker, D. D., & Hayes, A. F. (2007). Addressing moderated mediation hypotheses: theory, methods, and prescriptions. *Multivariate Behavioral Research*, 42, 185–227. <http://dx.doi.org/10.1080/00273170701341316>.
- Ratelle, C. F., Guay, F., Vallerand, R. J., Larose, S., & Senécal, C. (2007). Autonomous, controlled, and amotivated types of academic motivation: a person-oriented analysis. *Journal of Educational Psychology*, 99, 734–746. <http://dx.doi.org/10.1037/0022-0663.99.4.734>.
- Reeve, J. (2009). Why teachers adopt a controlling motivating style toward students and how they can become more autonomy supportive. *Educational Psychologist*, 44, 159–175. <http://dx.doi.org/10.1080/00461520903028990>.
- Rice, K. G., & Ashby, J. S. (2007). An efficient method for classifying perfectionists. *Journal of Counseling Psychology*, 54, 72–85. <http://dx.doi.org/10.1037/0022-0167.54.1.72>.
- Rice, K. G., & Slaney, R. B. (2002). Clusters of perfectionists: two studies of emotional adjustment and academic achievement. *Measurement and Evaluation in Counseling and Development*, 35(1), 35–48.
- Rice, K. G., Ashby, J. S., & Slaney, R. B. (2007). Perfectionism and the five factor model of personality. *Assessment*, 14, 385–398. <http://dx.doi.org/10.1177/1073191107303217>.

- Rice, K. G., Richardson, C. M., & Tueller, S. (2014). The short form of the revised almost perfect scale. *Journal of Personality Assessment*, 96, 368–379. <http://dx.doi.org/10.1080/00223891.2013.838172>.
- Ryan, R. M., & Connell, J. P. (1989). Perceived locus of causality and internalization: examining reasons for acting in two domains. *Journal of Personality and Social Psychology*, 57, 749–761. <http://dx.doi.org/10.1037/0022-3514.57.5.749>.
- Ryan, R. M., & Deci, E. L. (2000). Self-determination theory and the facilitation of intrinsic motivation, social development, and well-being. *American Psychologist*, 55, 68–78. <http://dx.doi.org/10.1037/0003-066X.55.1.68>.
- Salanova, M., Schaufeli, W., Martínez, I., & Bresó, E. (2010). How obstacles and facilitators predict academic performance: the mediating role of study burnout and engagement. *Anxiety, Stress & Coping*, 23, 53–70. <http://dx.doi.org/10.1080/10615800802609965>.
- Schaufeli, W. B., Martínez, I. M., Marques-Pinto, A., Salanova, M., & Bakker, A. B. (2002). Burnout and engagement in university students: a cross-national study. *Journal of Cross-Cultural Psychology*, 33, 464–481. <http://dx.doi.org/10.1177/0022022102033005003>.
- Schunk, D. H., Meece, J. L., & Pintrich, P. R. (2014). *Motivation in education: Theory, research and applications* (4rd ed.). Upper Saddle River, NJ: Merrill-Prentice Hall.
- Shim, H. (1995). *Effect of perfectionistic disposition on mental health: Base on factor analysis of perfectionism scale (master's dissertation)*. Seoul: Ewha Womens University.
- Shin, H., Puig, A., Lee, J., Lee, J. H., & Lee, S. M. (2011). Cultural validation of the maslach burnout inventory for Korean students. *Asia Pacific Educational Review*, 12, 633–639. <http://dx.doi.org/10.1007/s12564-011-9164-y>.
- Slaney, R. B., Rice, K. G., Mobley, M., & Trippi, J. (2001). The revised almost perfect scale. *Measurement and Evaluation in Counseling and Development*, 34, 130–145.
- Stoeber, J., & Eismann, U. (2007). Perfectionism in young musicians: relations with motivation, effort, achievement, and distress. *Personality and Individual Differences*, 43(8), 2182–2192.
- Stoeber, J., & Otto, K. (2006). Positive conceptions of perfectionism: approaches, evidence, challenges. *Personality and Social Psychology Review*, 10, 295–319. http://dx.doi.org/10.1207/s15327957pspr1004_2.
- Stoeber, J., Feast, A. R., & Hayward, J. A. (2009). Self-oriented and socially prescribed perfectionism: Differential relationships with intrinsic and extrinsic motivation and test anxiety. *Personality and Individual Differences*, 47, 423–428. <http://dx.doi.org/10.1016/j.paid.2009.04.014>.
- Suddarth, B. H., & Slaney, R. B. (2001). An investigation of the dimensions of perfectionism in college students. *Measurement and Evaluation in Counseling and Development*, 34, 157–165.
- Trumpeter, N., Watson, P. J., & O'Leary, B. J. (2006). Factors within multidimensional perfectionism scales: complexity of relationships with self-esteem, narcissism, self-control, and self-criticism. *Personality and Individual Differences*, 41, 849–860. <http://dx.doi.org/10.1016/j.paid.2006.03.014>.
- Vallerand, R. J., & Bissonnette, R. (1992). Intrinsic, extrinsic, and amotivational styles as predictors of behavior: a prospective study. *Journal of Personality*, 60, 599–620.
- Vallerand, R. J., & Ratelle, C. F. (2002). Intrinsic and extrinsic motivation: a hierarchical model. In E. L. Deci, & R. M. Ryan (Eds.), *Handbook of self-determination research* (pp. 37–69). Rochester, NY: The University of Rochester Press.
- Vallerand, R. J., Blais, M. R., Brière, N. M., & Pelletier, L. G. (1989). Construction et validation de l'échelle de motivation en éducation (EME). *Canadian Journal of Behavioural Science/Revue Canadienne des Sciences du Comportement*, 21, 323–349. <http://dx.doi.org/10.1037/h0079855>.
- Vallerand, R. J., Pelletier, L. G., Blais, M. R., Brière, N. M., Senécal, C., & Vallières, E. F. (1992). *Academic motivation scale (AMS-C 28) college version*. (Retrieved from http://er.uqam.ca/nobel/r26710/LRCS/echelles/Backup.11.05.05/EMEC_en.pdf).
- Vansteenkiste, M., Smeets, S., Soenens, B., Lens, W., Matos, L., & Deci, E. L. (2010). Autonomous and controlled regulation of performance-approach goals: Their relations to perfectionism and educational outcomes. *Motivation and Emotion*, 34, 333–353. <http://dx.doi.org/10.1007/s11031-010-9188-3>.
- Walls, T. A., & Little, T. D. (2005). Relations among personal agency, motivation, and school adjustment in early adolescence. *Journal of Educational Psychology*, 97, 23–31. <http://dx.doi.org/10.1037/0022-0663.97.1.23>.
- Weissman, A. N., & Beck, A. T. (1978). Development and validation of the Dysfunctional Attitude Scale: A preliminary investigation. *Paper presented at the annual meeting of the association for the advancement of behavior therapy, Chicago, Illinois*.
- Williams, G. C., Grow, V. M., Freedman, Z., Ryan, R. M., & Deci, E. L. (1996). Motivational predictors of weight loss and weight-loss maintenance. *Journal of Personality and Social Psychology*, 70, 115–126. <http://dx.doi.org/10.1037/0022-3514.70.1.115>.
- Zhang, Y., Gan, Y., & Chan, H. (2007). Perfectionism, academic burnout and engagement among Chinese college students: a structural equation modeling analysis. *Personality and Individual Differences*, 43, 1529–1540. <http://dx.doi.org/10.1016/j.paid.2007.04.010>.
- Zuckerman, M., Porac, J., Lathin, D., Smith, R., & Deci, E. L. (1978). On the importance of self-determination for intrinsically motivated behavior. *Personality and Social Psychology Bulletin*, 4, 443–446. <http://dx.doi.org/10.1177/014616727800400317>.